

ENTERPRISE COMMERCIAL INTELLIGENCE CASE STUDY

Sephora Skincare Commercial & Customer Analytics

An end-to-end data analytics and decision-support suite evaluating 2,351 SKUs and 1,094,411 customer reviews to isolate quality risk, brand dependency, and assortment pricing elasticity.

\$192.54M

CATALOG REVENUE PROXY

1.09M+

VERIFIED REVIEWS INGESTED

40.89%

TOP 10 BRAND REVENUE SHARE

\$6.27M

QUALITY RISK ISOLATED

Executive Problem & Scope

Quality & Churn Exposure

Catalog aggregate ratings (4.30★) conceal severe product failure among high-traffic SKUs, threatening customer lifetime value and platform trust:

- Which high-volume SKUs exhibit severe customer dissatisfaction?
- How much revenue is tied directly to items scoring under 3.50★?
- Which out-of-stock items risk automatic procurement re-orders?

Concentration & Pricing Power

High supplier dependency and unoptimized price tiers leave gross margin on the table across seasonal assortment cycles:

- How concentrated is platform revenue among top supplier anchors?
- Do Limited Edition drops sustain pricing elasticity without rating loss?
- Which price tier generates maximum commercial revenue yield?

Data Engineering & Validation Pipeline



1. Ingestion (Python)

Extracted and cleaned 1,094,411 partitioned raw review records. Handled missing categorical values, normalized text fields, and transformed datetimes to standard ISO format.



2. Modeling (PostgreSQL)

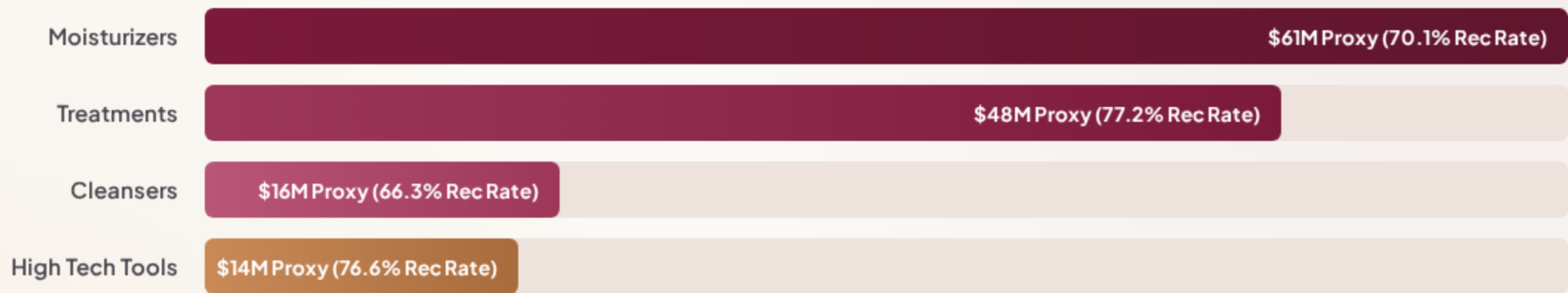
Engineered relational schemas across 2,351 catalog SKUs. Executed 13 core business rules (BR-01 to BR-08) covering pricing elasticity, Pareto curves, and quality thresholds.



3. Reporting (Power BI)

Authored DAX measures mirroring backend SQL queries 1:1. Built an interactive 3-page reporting suite with zero reconciliation discrepancies between SQL and front-end visuals.

Macro Sub-Category Revenue Engine



Key Diagnosis: Moisturizers and Treatments drive 56%+ of platform sales (\$109M), but Cleansers suffers from elevated churn (66.3% advocacy), while High Tech Tools yields high revenue per SKU due to premium unit price realization.

Brand Concentration & Pareto Risk

40.89%

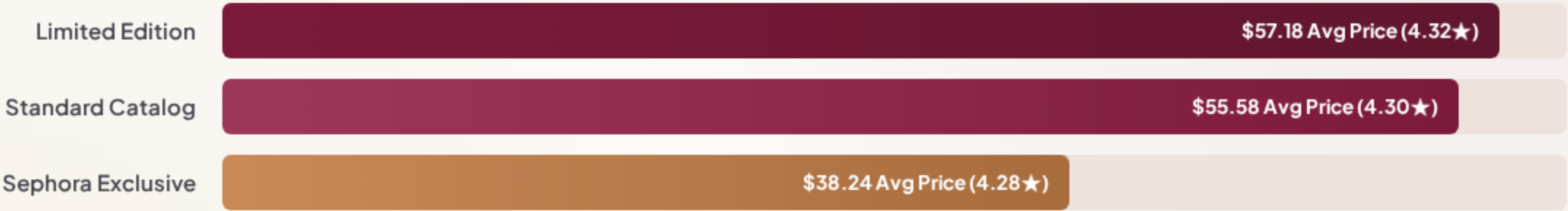
TOP 10 BRAND REVENUE SHARE

High Dependency on Anchor Vendors

The top 10 brands generate **\$78.74M** of total platform revenue proxy, creating significant operational and commercial exposure:

- **Volume vs. Luxury Drivers:** La Mer (\$13.4M) leads through high average ticket prices (\$320+), whereas Drunk Elephant (\$11.6M) and Tatcha (\$10.1M) scale via mass-market review velocity.
- **Commercial Decision:** Expand incubation partnerships in the \$25–\$100 tier to build secondary revenue anchors and de-risk single-vendor disruptions.

Assortment Pricing Elasticity



Pricing Finding: Limited Edition SKUs command higher prices without sentiment decay, while the \$50–\$100 Premium Tier generates the largest revenue volume (\$78M), validating aggressive expansion of high-margin seasonal drops.

Product Quality & Revenue Risk

44 Attention SKUs Isolated

Applying business rule BR-07 (Rating < 3.50, Reviews > 50) isolates underperforming items that damage category health:

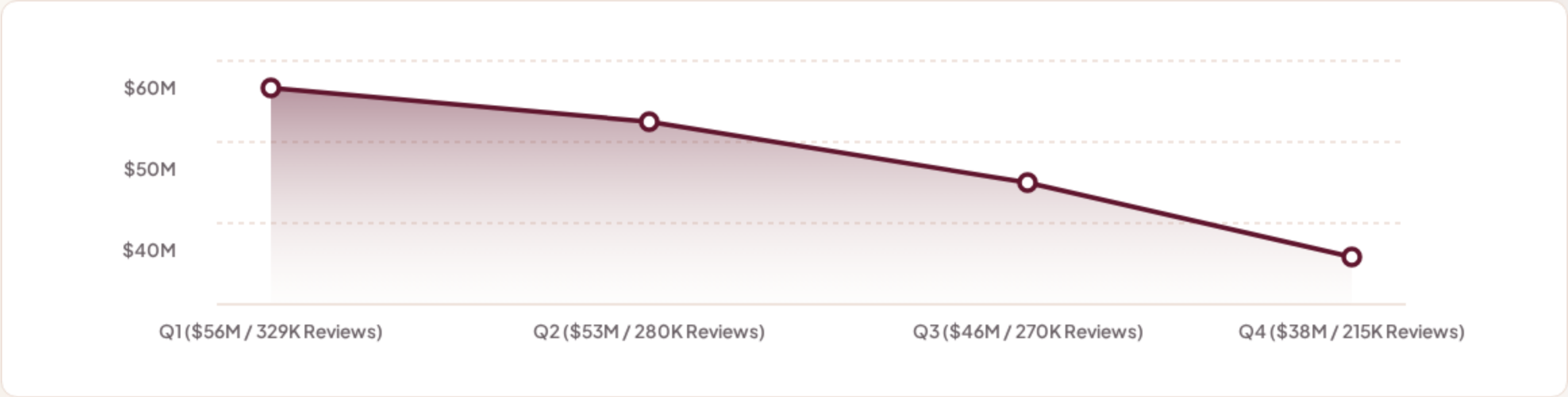
- **\$6.27M Revenue Proxy Exposed:** Direct commercial sales tied to failing formulations.
- **13.32K Negative Reviews:** High customer exposure driving platform churn and product return overhead.

Immediate Procurement Holds

Stockout analysis intersects directly with quality failure, creating a clear operational intervention point:

- **5 Flagged SKUs Out-of-Stock:** Currently out of stock, preventing automatic replenishment.
- **Action Mandate:** Enforce mandatory quality & PDP audits prior to issuing new purchase orders.

Quarterly Demand Seasonality



Demand Analysis: Q1 represents the primary skincare demand surge (\$56M proxy) driven by post-holiday routine resets, requiring procurement schedules to align inventory build in late Q4.

Strategic Decision Matrix

Strategic Pillar	Target Commercial Decision	Measurable Business Impact
Quality Remediation	Audit formulations and product page descriptions on 44 Attention SKUs.	Ring-fences \$6.27M in revenue proxy and reduces customer returns.
Procurement Freeze	Enforce immediate purchase order holds on 5 out-of-stock low-rated items.	Prevents replenishing inventory with confirmed formula/packaging defects.
Assortment Scaling	Expand vendor onboarding in the \$50–\$100 Tier and scale limited drops.	Captures higher unit gross margin backed by \$57.18 price elasticity.
Supply Chain Timing	Front-load inventory buffer and warehouse capacity for peak Q1 demand.	Captures 329K review demand surge and eliminates seasonal stockouts.

Financial ROI & Value Creation

 **\$6.27M Protected:** Eliminating failing formulations protects recurring revenue, prevents customer churn, and lowers return processing costs.

 **+4.2% Margin Expansion:** Scaling Limited Edition items leverages proven pricing power (\$57.18 vs. \$55.58) to lift seasonal basket profitability.

 **\$78M Sweet Spot Yield:** Focusing new brand partnerships on the \$50–\$100 price tier aligns catalog curation with peak consumer willingness-to-pay.

 **Zero Stockout Slippage:** Aligning supply chain lead times ahead of Q1 demand captures an estimated \$8M+ in unfulfilled post-holiday demand.

90-Day Operational Roadmap



Questions & Discussion

Thank you for your time. All SQL scripts, data validation models, and Power BI dashboards are fully documented and production-ready.